

Making a Wired Printer Wireless using a Raspberry Pi - Full Setup Guide

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In this guide, you will be led through the process of configuring a Raspberry Pi to transform a wired USB printer into a WiFi-enabled printer, which you can print to wirelessly from any device on your network.

The guide covers the three major steps needed: configuring your Pi, setting up the printing software, and connecting to it from the computer you want to print from. It's intended for those with little technical experience, but should be usable for anyone with the parts required.

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Setting up the Pi

In this section, you'll set up your Raspberry Pi and prepare it for use. Before you begin, gather the required parts, listed below

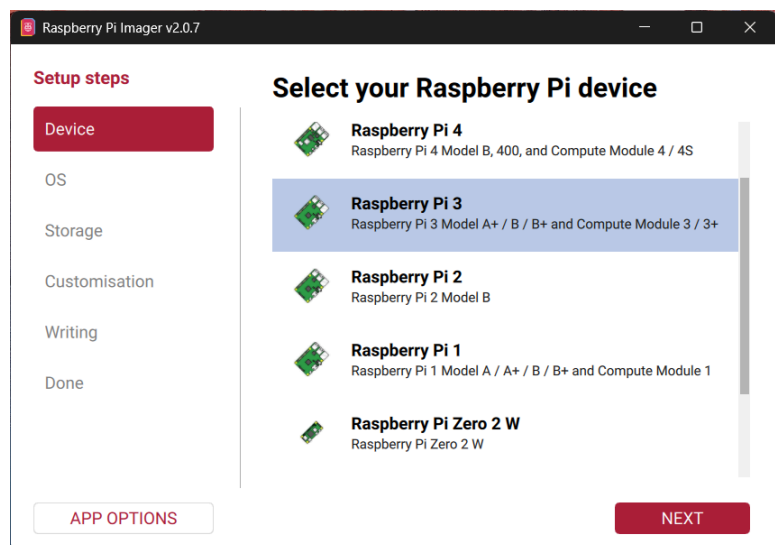
What you'll need:

- ☐ Raspberry Pi 3 B or Zero W or better
- ☐ Micro-SD card with full-size SD adapter
- ☐ Appropriate cables
 - ☐ 5V, 2.5A Micro-USB power supply (2.5A or greater required)
 - ☐ HDMI cable (If you don't have access to your router settings)
 - ☐ Micro-USB to USB-A adapter (Raspberry Pi Zero W, W2)
 - ☐ Mini-HDMI to HDMI adapter (HDMI cable & Raspberry Pi Zero W, W2)

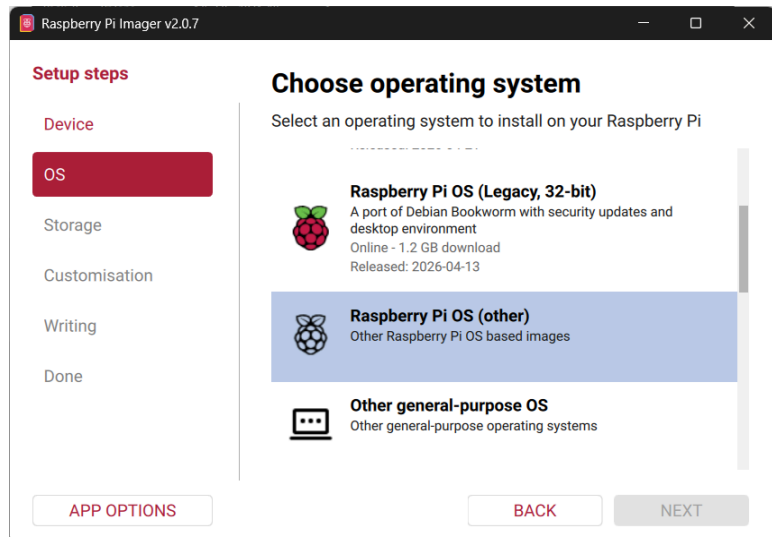
1. Preparing the SD Card

The [Raspberry Pi Imager](#) is a software to prepare your SD card for use with your Raspberry Pi. Install Raspberry Pi Imager and insert the SD card into your computer, then launch the Imager program.

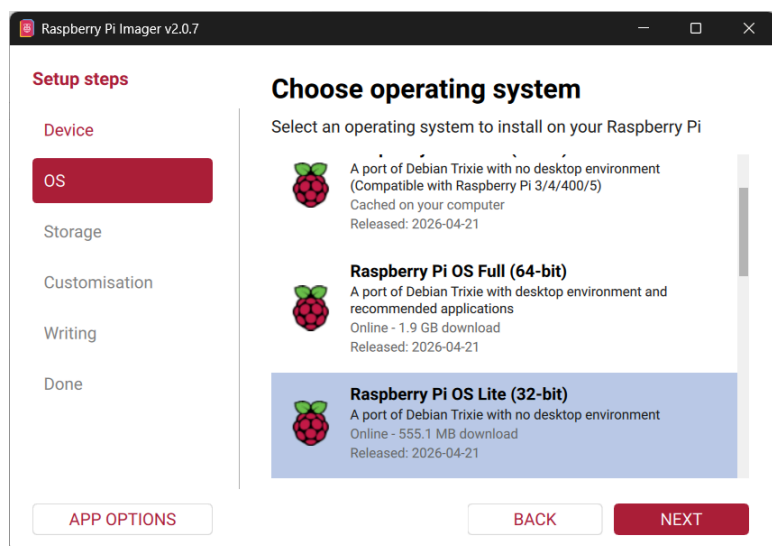
After launch, you will see the “Device” screen. Select the device that describes your Pi. For example, if you have a Raspberry Pi Model 3B+, select “Raspberry Pi 3” as shown. Press “Next”.



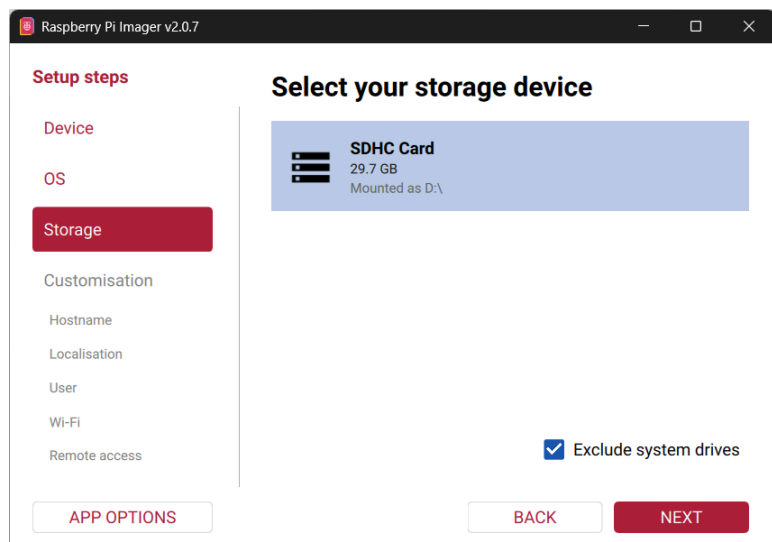
Now we will choose the correct operating system to run on your Pi. Start by selecting Raspberry Pi OS (other). This will bring up a second menu.



Under the second menu, select “Raspberry Pi OS Lite (32-bit)”. This OS will run on any required Pi model, but if you know your device supports 64-bit, you may choose “Raspberry Pi OS Lite (64-bit)”

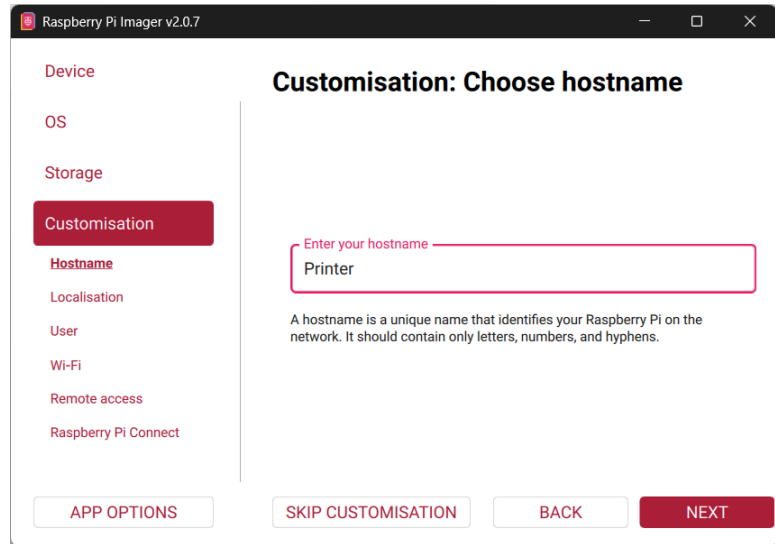


After selecting your OS, you will be prompted to select your storage device to store the operating system and data. Make sure your SD card is inserted and choose it from the menu, typically listed as “SDHC Card” or similar.



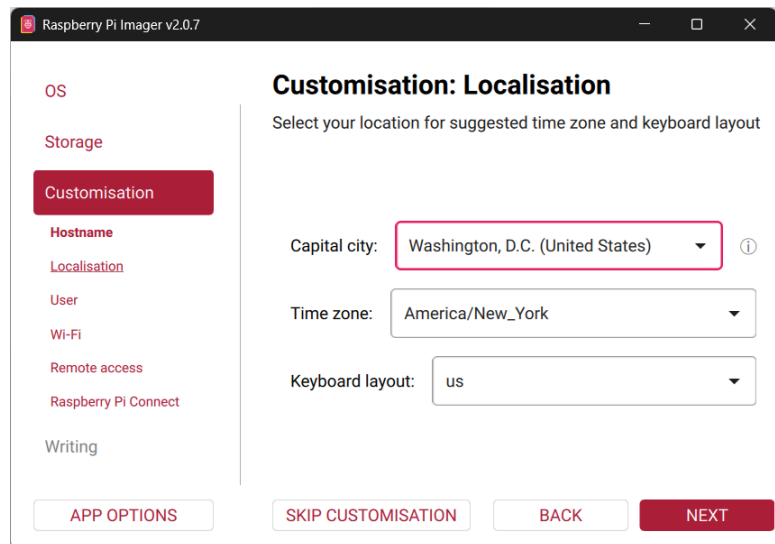
Ensure “Exclude system drives” is **checked**.

The “Customization” menu will ask you a few questions to set up your system. The first is your hostname, which is a name that your Pi uses to identify itself on your network. Choose a memorable name with no spaces or special characters, such as “PrinterPi”.



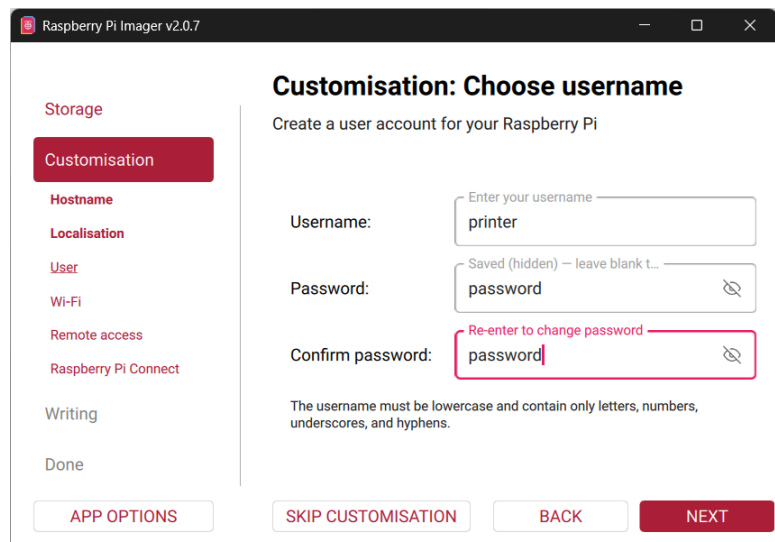
The screenshot shows the 'Customisation: Choose hostname' screen in the Raspberry Pi Imager v2.0.7. On the left, a sidebar lists various settings: Device, OS, Storage, Customisation (highlighted), Hostname, Localisation, User, Wi-Fi, Remote access, and Raspberry Pi Connect. The main area is titled 'Customisation: Choose hostname' and features a text input field labeled 'Enter your hostname' with the value 'Printer' entered. Below the input field, a note states: 'A hostname is a unique name that identifies your Raspberry Pi on the network. It should contain only letters, numbers, and hyphens.' At the bottom, there are four buttons: 'APP OPTIONS', 'SKIP CUSTOMISATION', 'BACK', and 'NEXT'.

Next is information about your location and time zone. Be sure to choose the correct capital city for your country, as it is used to determine the proper Wi-Fi protocol to use. None of this information is stored outside of your computer and is not privacy sensitive.



The screenshot shows the 'Customisation: Localisation' screen in the Raspberry Pi Imager v2.0.7. The sidebar on the left is the same as the previous screen, with 'Localisation' highlighted. The main area is titled 'Customisation: Localisation' and includes the instruction 'Select your location for suggested time zone and keyboard layout'. It contains three dropdown menus: 'Capital city' (set to 'Washington, D.C. (United States)'), 'Time zone' (set to 'America/New_York'), and 'Keyboard layout' (set to 'us'). At the bottom, there are four buttons: 'APP OPTIONS', 'SKIP CUSTOMISATION', 'BACK', and 'NEXT'.

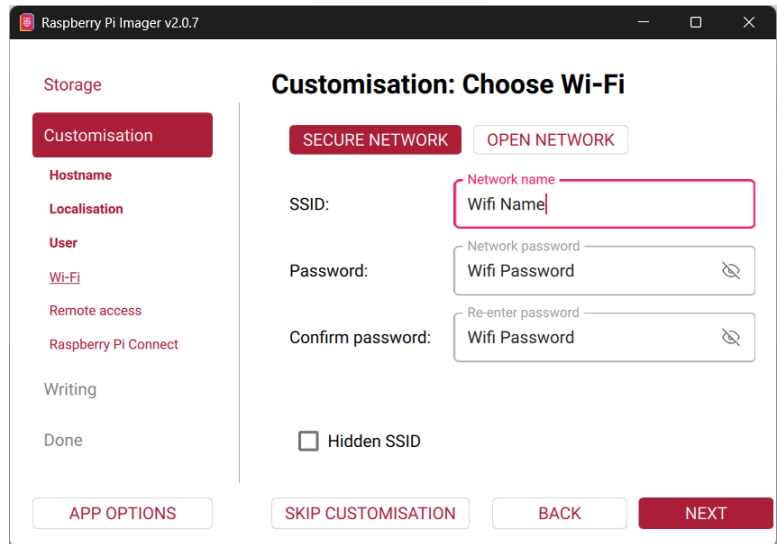
Now you will set up a user account. Choose a username and password you will remember and write them down, as **you cannot recover them later**.



The screenshot shows the 'Customisation: Choose username' screen in the Raspberry Pi Imager v2.0.7. The sidebar on the left is the same as the previous screens, with 'User' highlighted. The main area is titled 'Customisation: Choose username' and includes the instruction 'Create a user account for your Raspberry Pi'. It contains three input fields: 'Username' (with 'printer' entered), 'Password' (with 'password' entered and a 'Saved (hidden) — leave blank' note), and 'Confirm password' (with 'password' entered). A note at the bottom states: 'The username must be lowercase and contain only letters, numbers, underscores, and hyphens.' At the bottom, there are four buttons: 'APP OPTIONS', 'SKIP CUSTOMISATION', 'BACK', and 'NEXT'.

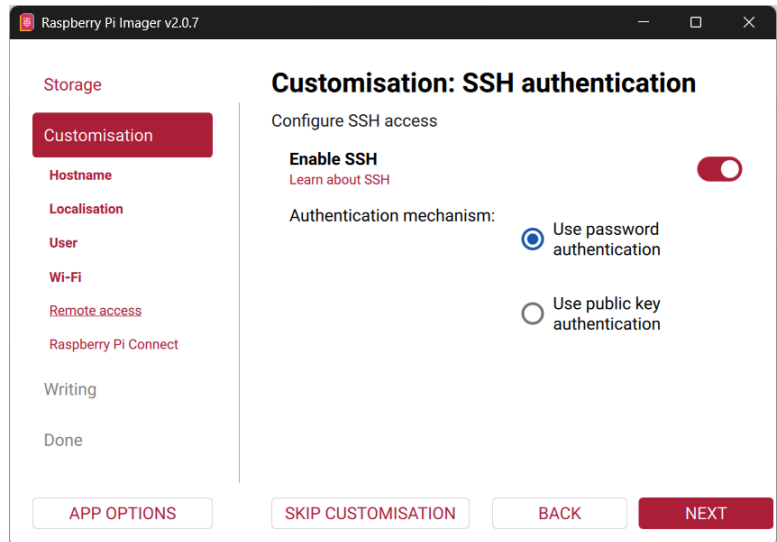
This will be used to log in and run certain commands.

To connect your Pi to your Wi-Fi network, you can provide connection details in advance. Enter your Wi-Fi name (SSID) and the password you use to connect. If your network doesn't require a password, choose "OPEN NETWORK".



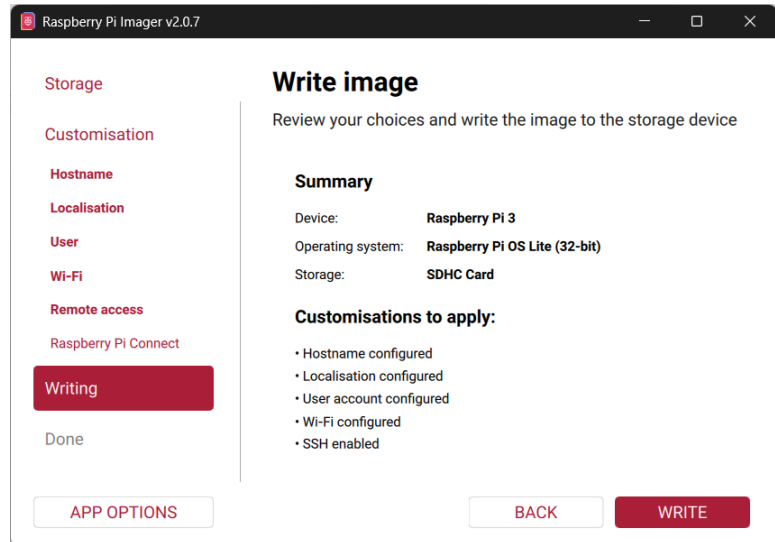
The screenshot shows the 'Customisation: Choose Wi-Fi' screen in the Raspberry Pi Imager v2.0.7 application. On the left, a sidebar lists various customisation options: Storage, Customisation (selected), Hostname, Localisation, User, Wi-Fi (highlighted in red), Remote access, Raspberry Pi Connect, Writing, and Done. The main area is titled 'Customisation: Choose Wi-Fi' and features two tabs: 'SECURE NETWORK' (selected) and 'OPEN NETWORK'. Below the tabs, there are three input fields: 'SSID:' with a text box containing 'Wifi Name', 'Password:' with a text box containing 'Wifi Password' and a toggle icon, and 'Confirm password:' with a text box containing 'Wifi Password' and a toggle icon. A 'Hidden SSID' checkbox is located below the password fields. At the bottom, there are four buttons: 'APP OPTIONS', 'SKIP CUSTOMISATION', 'BACK', and 'NEXT'.

Finally, you must enable SSH, a way to control your Pi from your typical computer. Ensure the slider is on to enable SSH and "Use password authentication" is selected. Skip through the Raspberry Pi Connect screen and leave the feature disabled.

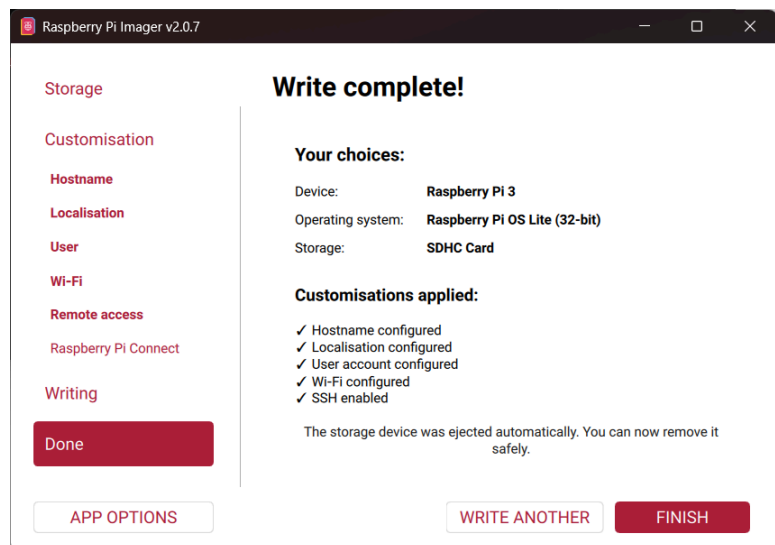


The screenshot shows the 'Customisation: SSH authentication' screen in the Raspberry Pi Imager v2.0.7 application. The sidebar on the left is identical to the previous screen, with 'Wi-Fi' highlighted. The main area is titled 'Customisation: SSH authentication' and has a subtitle 'Configure SSH access'. It features a toggle switch for 'Enable SSH' which is turned on. Below this, there is a link 'Learn about SSH'. Under the heading 'Authentication mechanism:', there are two radio button options: 'Use password authentication' (selected) and 'Use public key authentication'. At the bottom, there are four buttons: 'APP OPTIONS', 'SKIP CUSTOMISATION', 'BACK', and 'NEXT'.

Congrats! You're now ready to write your OS and settings to your SD card. Ensure the "Customisations to apply" match the screenshot and select "Write". A popover menu will appear, warning you that any data on the SD card will be overwritten. If you have any important data on your card that you must save, STOP NOW and move it to your computer. After waiting a few seconds, the menu will allow you to confirm your decision and will begin writing your SD card.

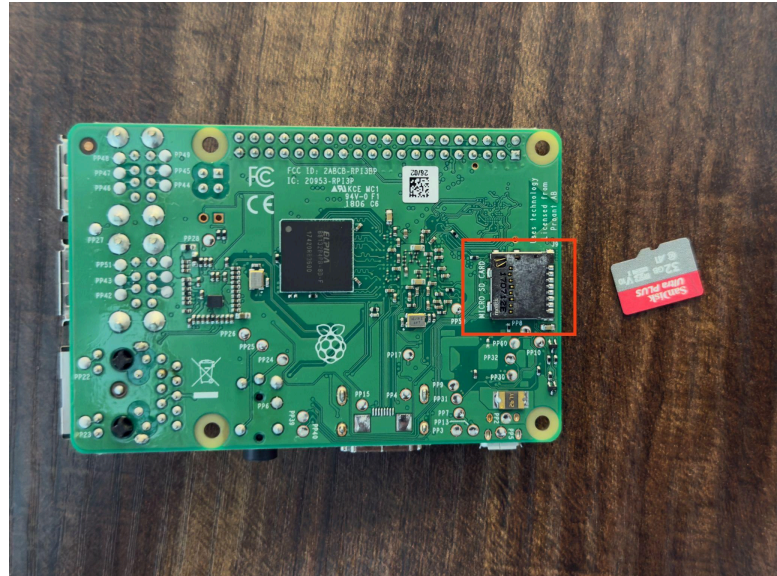


This process will take a few minutes, depending on the speed of the card. Let both the write and verify steps complete without interruption.



2. First time booting of your Pi

You are now ready to boot up your Pi for the first time! Now, depending on if you have access to your router settings and can see device IPs through the management tab of your router, this step is going to be slightly different. If you do not have the ability to view all devices connected to your network, you will need to plug your Pi into a screen using an HDMI cable.



First, insert the micro-SD card into the slot on the bottom of the Pi, boxed in the image. Then, plug in the power supply. The first boot process will take around 3-5 minutes. To confirm that it is finished booting, wait for the green LED on the Pi to stop flashing for at least 10 seconds.

If your Pi is connected with HDMI:

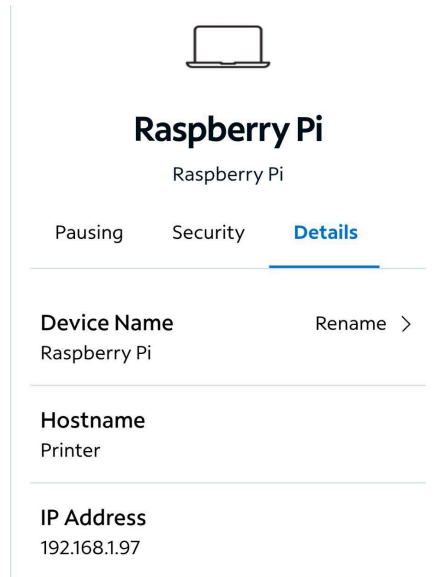
If you have your Pi plugged into a screen, it will show information similar to the image below when the boot process is complete.

The screen shows two IP addresses, and you need the second IP. An example IP is boxed in the image below. Record the IP address as you will need it for future steps!

```
Debian GNU/Linux 13 Printer tty1
My IP address is 127.0.1.1 fd00:2e67:be9b:b7cb:ba27:ebff:fe82:3d15
Printer login:
Debian GNU/Linux 13 Printer tty1
My IP address is 192.168.1.30 fd00:2e67:be9b:b7cb:ba27:ebff:fe82:3d15
Printer login: [ 34.549097] sh[984]: Completed socket interaction for boot stage config
[ 56.093062] sh[989]: Completed socket interaction for boot stage final
```


If you have a device list:

If you have access to the device list on your local network, for example through your router's management page, locate the Pi there. When you obtain the IP address, record it somewhere since you will need this in later steps.



Your Pi is now ready to be turned into a server for your printer!

Setting up the Printer

Now that your Pi is prepared, the printer software, drivers, and connection need to be set up. This section begins with installing and configuring the printing software on the Pi before finishing the connection on your computer.

1. SSH-ing into your Pi

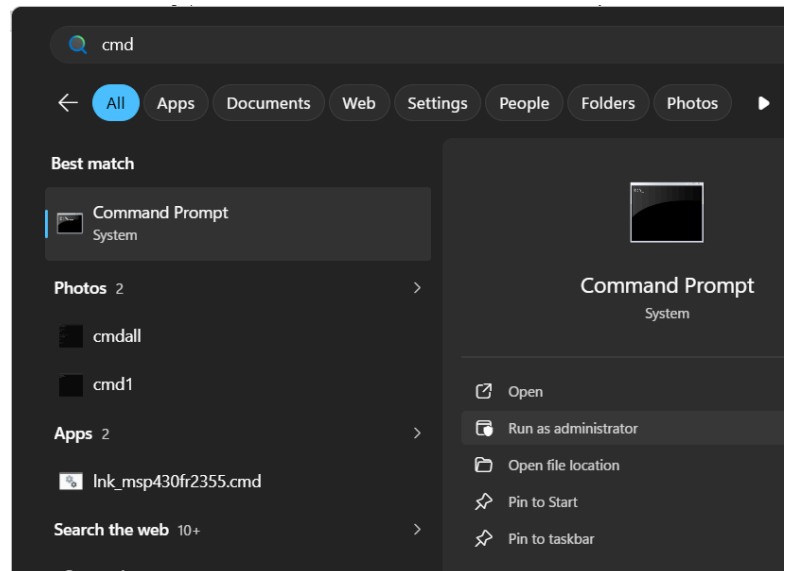
You will now need to run some commands in your command prompt to set things up. When running commands, type them exactly as written, and replace bracketed text without including the square brackets.

To connect to your Raspberry Pi and set up its software, you're going to connect to it through a Secure Shell, or SSH. To do this, open your command prompt as an administrator by pressing the WINDOWS key then typing in the search bar "cmd" then clicking the highlighted "Run as administrator" as seen, and run the command `ssh`

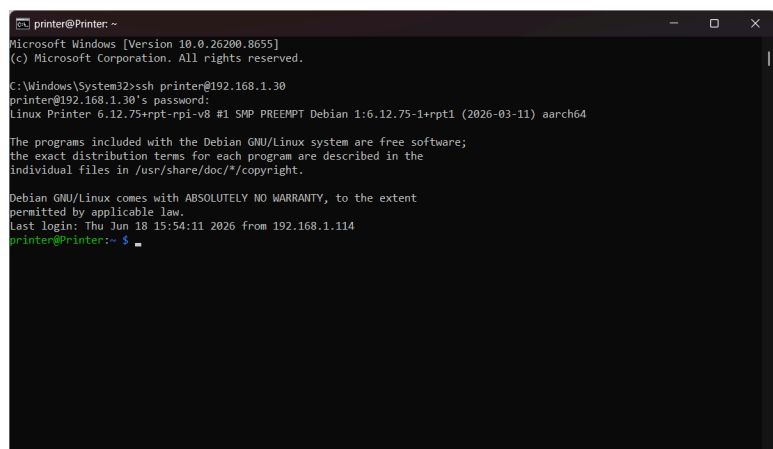
`[Pi_username]@[IP_address]`.

Replace Pi_username with the name of the user account that you created in Raspberry Pi

Imager, and IP_address with the IP that you found at first boot. Remember to not include the brackets in your command.



Please note, when it asks for your password, use the password you set for your Raspberry Pi when setting it up. You will not be able to see it while you type it. If you are asked if you want to continue with the

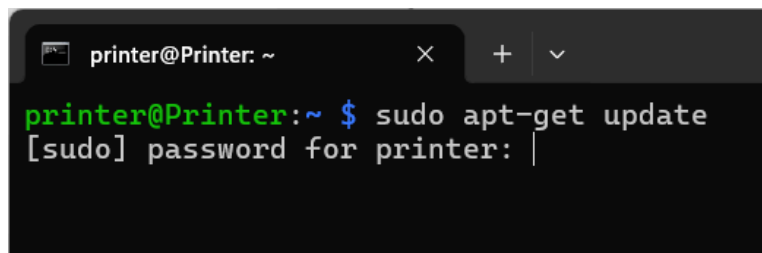


connection due to security reasons, type 'yes'.

2. Setting up the Software

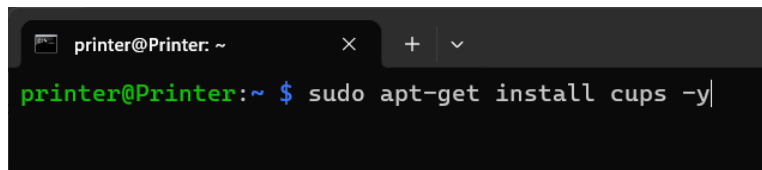
The first command will update a software called APT, which comes pre-installed on your Raspberry Pi's operating system. APT manages the installation of other software on your system, and updating it ensures that it has access to the latest software versions when you use it.

To update APT, run `sudo apt-get update`. It may ask you for your password (as shown below), and you'll have to enter it. Once again, you will not be able to see your password as you type it. Next, you will be installing the printing system, called CUPS, using APT. This software will allow you to access your printer over Wi-Fi.



```
printer@Printer: ~  
printer@Printer:~ $ sudo apt-get update  
[sudo] password for printer: |
```

To install CUPS, run the command `sudo apt-get install cups -y`.



```
printer@Printer: ~  
printer@Printer:~ $ sudo apt-get install cups -y|
```

3. Setting up the Printer System

You are almost done with the process of entering setup commands. The next set of commands will set up permissions to allow you to access the printer from a remote computer, such as the one you are using right now.

First, run `sudo usermod -aG lpadmin [pi_username]`, which allows you to modify printers and their connections to the Raspberry Pi. Remember that `pi_username` is the name you entered when creating your user.

```
printer@Printer: ~  
printer@Printer:~ $ sudo usermod -aG lpadmin printer_
```

Next, you're going to enable the sharing and remote access of your printers connected to the Pi by using `sudo cupsctl --share-printers --remote-any`.

```
printer@Printer: ~  
printer@Printer:~ $ sudo cupsctl --share-printers --remote-any|
```

Almost done! You need to enable the web interface for the printers so you can more easily add new printers and access admin settings. The command to run is `sudo cupsctl WebInterface=yes`.

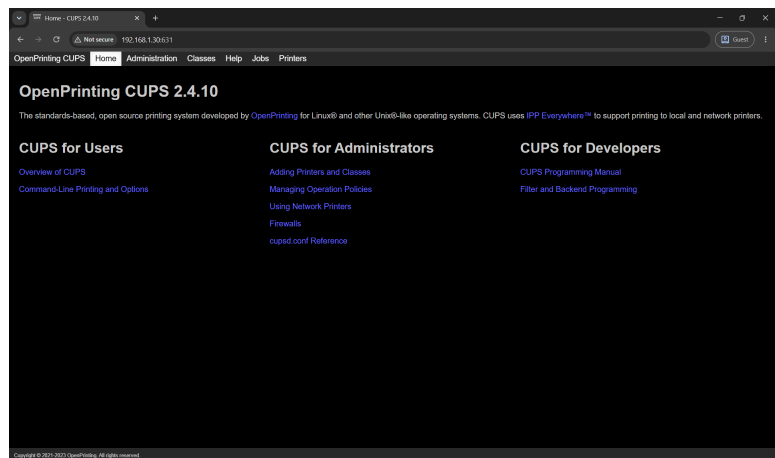
```
printer@Printer: ~  
printer@Printer:~ $ sudo cupsctl WebInterface=yes_
```

To restart the printing service and apply your changes, run `sudo systemctl restart cups`. You'll be running this again later and should run it after you make any changes to CUPS through the command prompt.

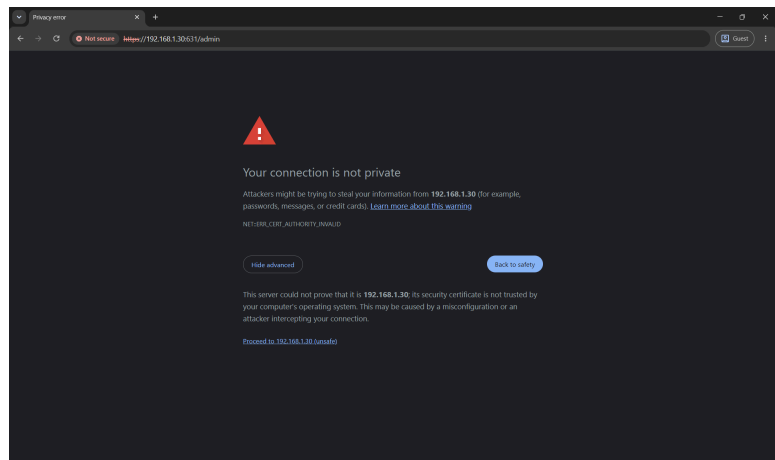
```
printer@Printer: ~  
printer@Printer:~ $ sudo systemctl restart cups_
```

You are done with typing commands for a little bit and now need to open your web browser of choice, but make sure you keep your command prompt open in the background.

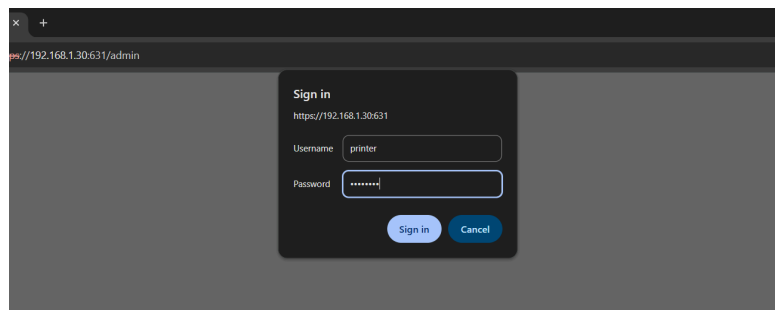
Then, type `[IP_address]:631` into the address bar.



From the above page, you'll go to the "Administration" tab at the top bar. If you see an error about an upgrade being required, just wait a minute and it'll automatically bring you to the administration page. At this point, you'll probably see a security warning, but this is a false alert from your browser and is safe to ignore. Click on "Advanced". then the link at the bottom that says "Proceed".



Once you proceed, you'll see a prompt to log in. Use the credentials for your Raspberry Pi user.

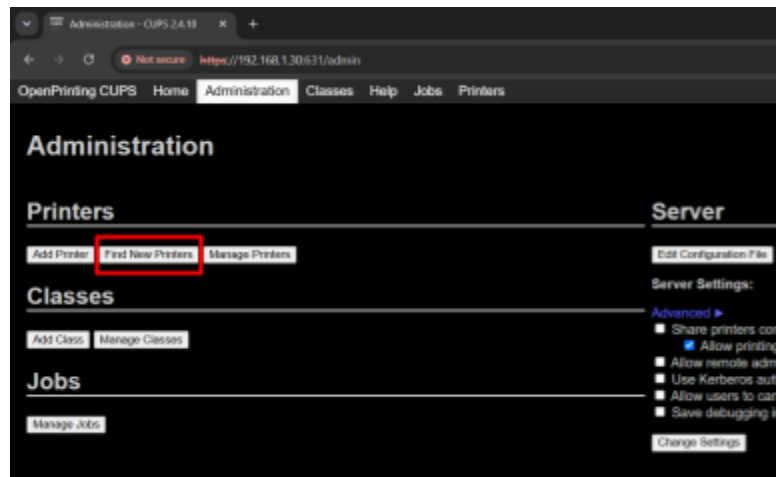


Once you log in, you'll be taken to the administration page where you can add the printer you have plugged into your Raspberry Pi.

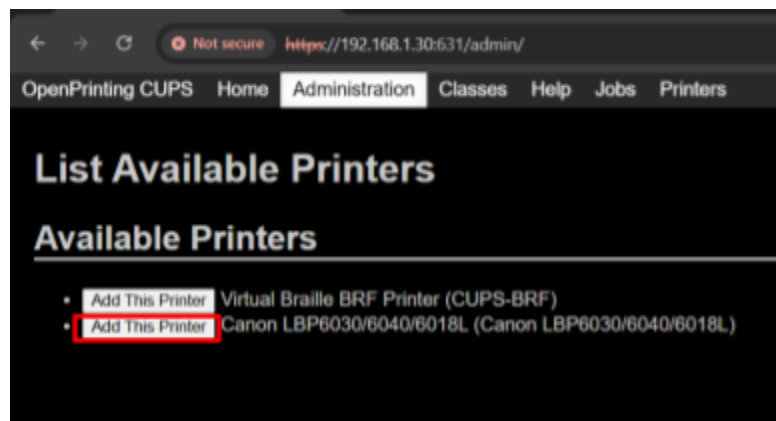
Make sure the printer is plugged into any of the USB ports of the Pi and is powered on from this point forward.

4. Adding the Printer Using the Web Interface

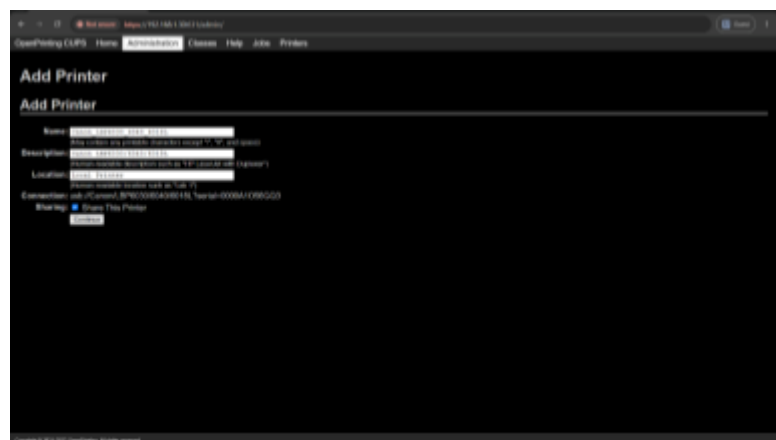
In the Administration tab,
select “Find New Printers”



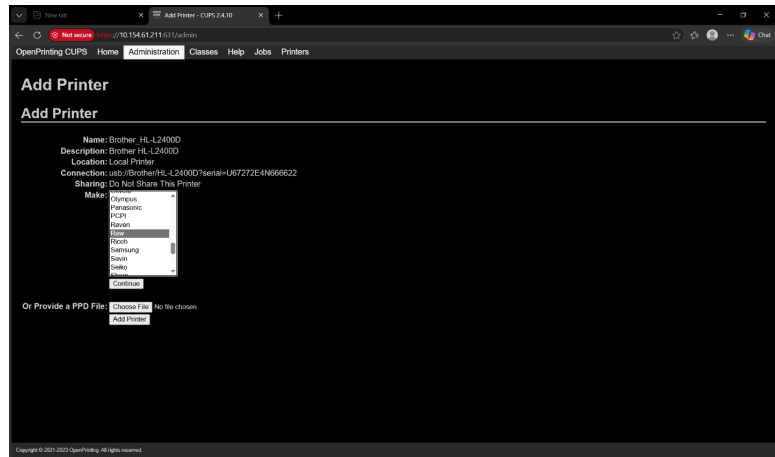
Next, select “Add This
Printer” for whichever
printer is plugged into your
Raspberry Pi. In the
example, it is the Canon
LBP6030 printer.



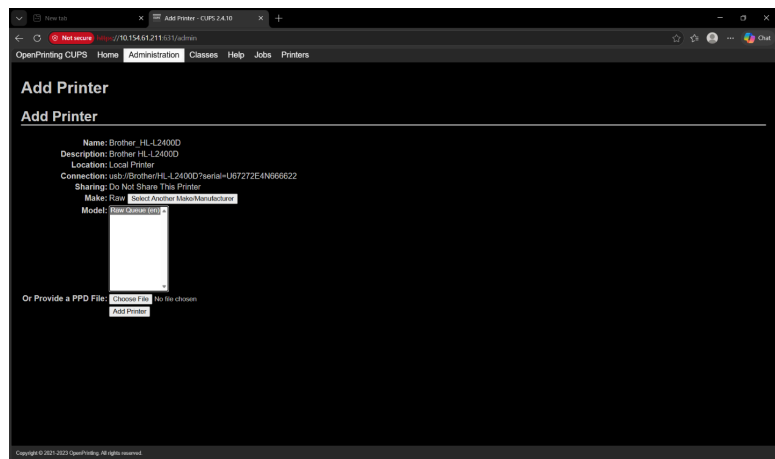
You can now set a custom
printer name and
description if you so desire,
but this is not required and
you can choose to continue
to the next step without
entering anything.
Checking “Share This
Printer” is also optional
because you will be setting
this later through a command.



On the next page, you'll be asked to select the printer driver. You'll use a "raw queue", which means the data sent to the printer is exactly what the Pi receives from your computer, without any extra processing. This will mean you'll need to install a driver onto the client computer you are printing from so that the printer supports the data it is sent, but we'll handle that later.

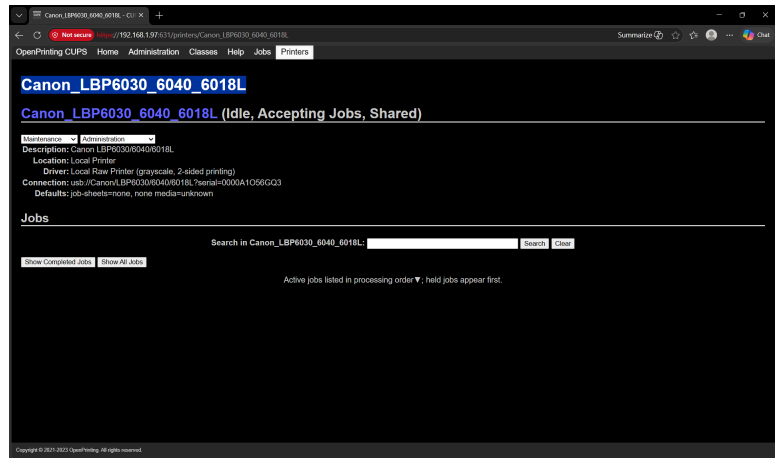


To select a raw queue as the driver, find and choose "Raw" from the "Make" dropdown menu, as shown in the first image. Click the continue button. Then, you will see a "Model" dropdown, as in the second image. In this menu, choose "Raw Queue (en)".



To save the printer and driver, click the "Add Printer" button at the bottom of the options.

Now you'll navigate to the Printers tab in the top bar and select the queue for the printer you just added. Highlight and copy the bold printer name at the top of the page once the queue has been selected. This will be the printer name you set earlier if you chose a custom one, or the default for your type of printer if you did not.



Switch back to your SSH command prompt. Using the name you just copied, run the command `sudo lpadmin -p [printer_name] -o printer-is-shared=true`. This will make sure the printer is accessible remotely.

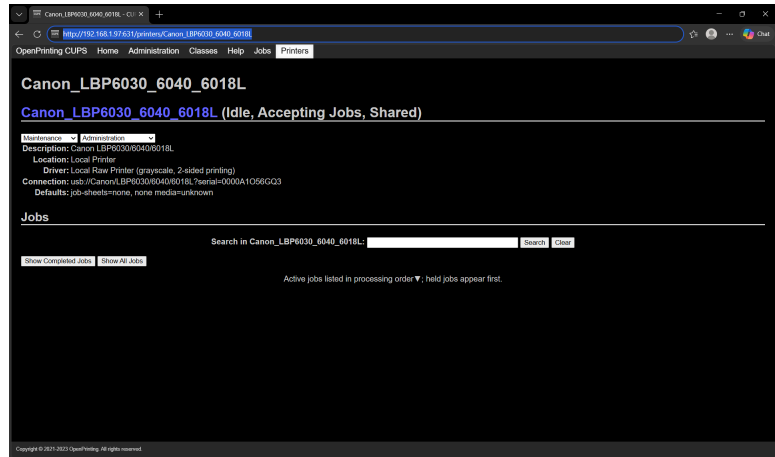
```
printer@Printer: ~  
printer@Printer:~$ sudo lpadmin -p Canon_LBP6030_6040_6018L -o printer-is-shared=true
```

Run the restart command again, and you will be done with the setup! You can now add the printer to your local computer with the steps below.

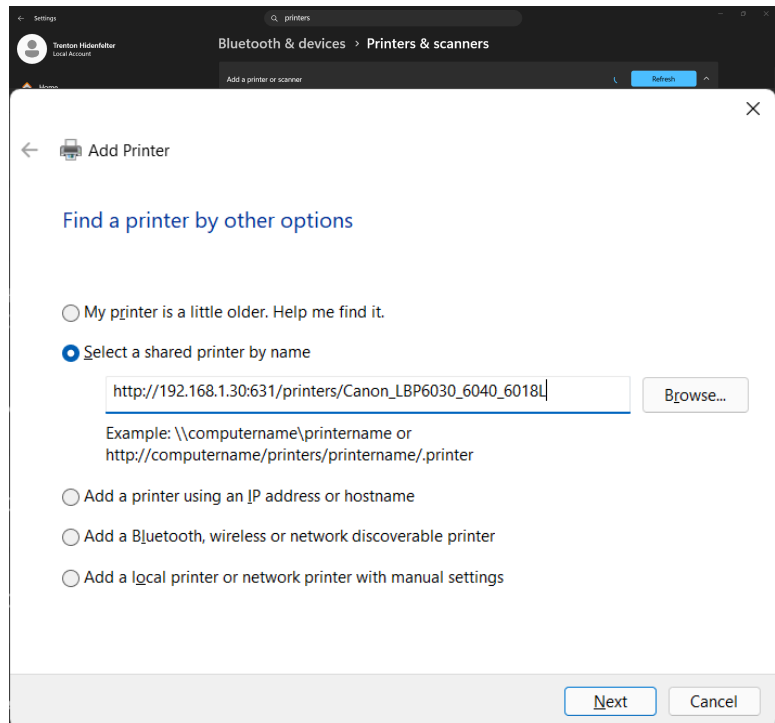
```
printer@Printer: ~  
printer@Printer:~$ sudo systemctl restart cups
```

5. Adding the Printer to your Computer

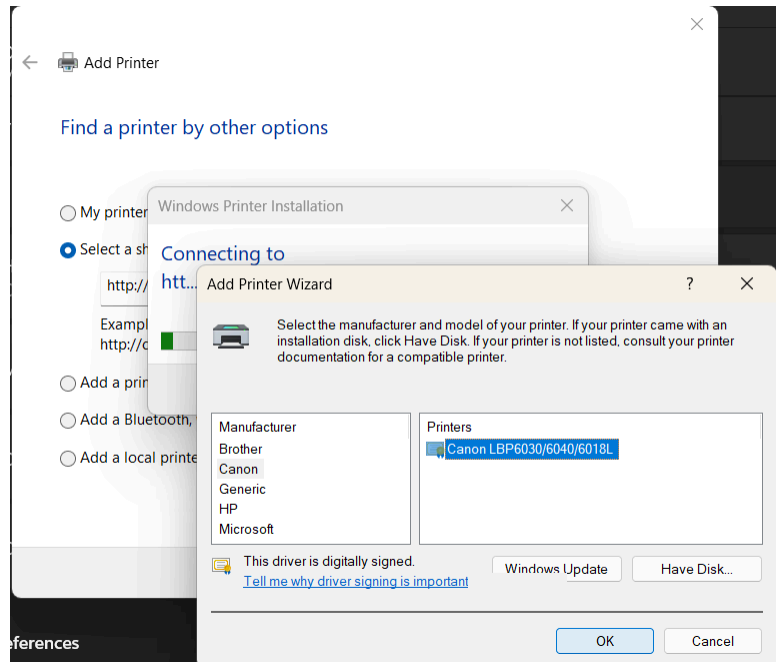
While you have your printer selected in the “Printers” tab as shown, copy the URL in the address bar, highlighted in the image.



Next, open your settings and navigate to “Printers & scanners” using the search bar, select “Add a printer or scanner”, then “Add a Printer Manually”, then “Select a shared printer by name” and paste in the URL you just copied. Change where it says https:// to http:// (no ‘s’) if it is not already.



Next you'll be asked to select a driver. Install the driver for your printer from the manufacturer's website if you don't already have it installed. Find and select the driver for your printer if the driver is displayed in the list. If it is not in the list, choose "Have Disk" and find the driver file on your disk. You can find the driver for your printer on the support pages for your printer.



Select OK and you'll be all set up to use your wired printer wirelessly from your computer! Make sure your Pi and printer are powered on and connected whenever you plan to print.